Aligning a Korean Chatbot with Human Feedback:

From Supervised Fine-Tuning to Policy Optimization

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Abstract

Instruction-following chatbots in Korean are difficult to build because the language relies on rich contextual cues and a layered honorific system that standard training struggles to capture. This paper presents a three-stage pipeline—Supervised Fine-Tuning (SFT), Reward Modeling (RM), and Proximal Policy Optimization (PPO)—for aligning a Korean chatbot with human preferences under the Reinforcement Learning from Human Feedback (RLHF) framework. We also apply data augmentation at the SFT stage using Word2Vec synonym replacement, beam search paraphrasing, and noise injection. Experiments on the KoChatGPT dataset show that the PPO-trained model achieves the lowest perplexity (7.39) and competitive diversity scores across five automatic metrics, suggesting that even a modestly-resourced RLHF pipeline can meaningfully improve response quality for Korean.

1 Introduction

Large Language Models (LLMs) can generate fluent text, but fluency alone does not mean a model will respond the way a user actually wants. A model trained to predict the next word optimizes for statistical patterns, not usefulness—a gap commonly referred to as the *alignment problem*.

Reinforcement Learning from Human Feedback (RLHF) is one of the more practical approaches to closing this gap. InstructGPT [2] showed that RLHF meaningfully improved how well a model's outputs matched user intent. However, most RLHF resources are built around English. Korean poses additional difficulties: its layered honorific system and strong dependence on implicit context make it harder to align through standard supervised training alone [5].

This paper builds a full Korean RLHF pipeline (SFT $\rightarrow$ RM $\rightarrow$ PPO) using the open-source KoChatGPT dataset [1]. We train three model variants and evaluate them on five automatic metrics, with Word2Vec-based multi-strategy augmentation applied at the SFT stage to test whether more varied training data improves alignment. The main contributions are as follows:

- A complete Korean RLHF pipeline (SFT → RM → PPO) built on open-source tools and the KoChatGPT dataset.
- A Word2Vec-based multi-strategy augmentation method evaluated at the SFT stage.
- Quantitative comparison of three model variants across five metrics, with qualitative analysis of response quality.

2 Dataset

2.1 KoChatGPT Dataset

The KoChatGPT dataset [1] is a Korean instruction-following dataset split into three files, one per training stage:

- `kochatgpt_1_SFT.jsonl`: Prompt and completion pairs used to train the SFT model.
- `kochatgpt_1_RM.jsonl`: Each entry contains a prompt along with three ranked candidate responses, which are used to train the Reward Model.
- `kochatgpt_1_PPO.jsonl`: A set of prompts used during the PPO training stage.

This structure mirrors the data design used in InstructGPT [2].

2.2 Data Augmentation for SFT

Korean instruction-following datasets are relatively small, so data augmentation is applied to the SFT training split to expand variety. Three strategies are combined:

1. **Word2Vec synonym replacement**: Words in the completion sentences are replaced with semantically similar words identified through a pretrained Korean Word2Vec model. This preserves the general meaning while introducing surface-level variation.
2. **Beam search paraphrasing**: After the SFT model is trained, it is used to generate alternative completions for the same prompts via beam search. This produces structurally varied responses that differ from the original completions while preserving the general meaning.
3. **Noise injection**: Small perturbations (e.g., random word deletion or swap) are applied to some training examples. This encourages the model to rely on semantic content rather than surface-level patterns.

The resulting augmented dataset is used to train SFT-aug; results are compared against SFT-base, which is trained on the original data only.

3 Methodology

The training pipeline follows the three-stage RLHF approach of prior work [2, 4], as illustrated in Figure 1.

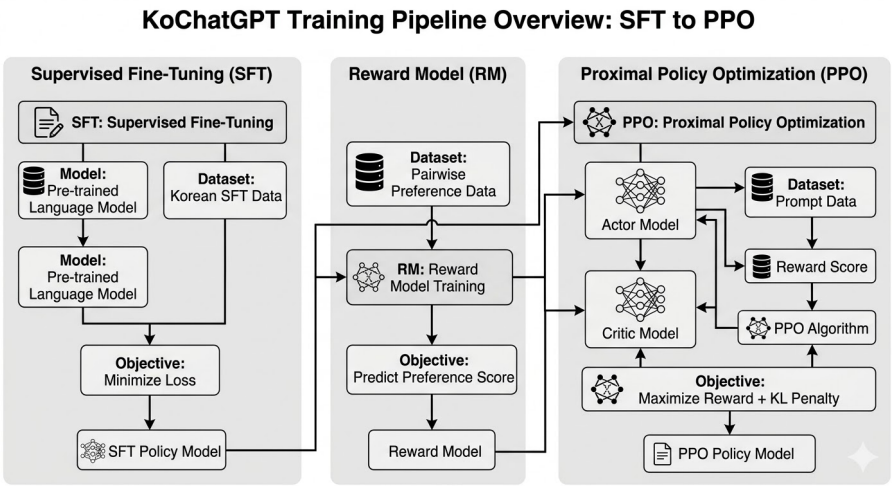


Figure 1: Overview of the three-stage RLHF training pipeline: SFT → Reward Modeling → PPO.

3.1 Stage 1: Supervised Fine-Tuning (SFT)

KoGPT-2 is fine-tuned on the SFT split of KoChatGPT using cross-entropy loss, teaching the model to generate appropriate responses to Korean prompts. Two versions are trained: **SFT-base** on the original data, and **SFT-aug** on the augmented version described in Section 2.

3.2 Stage 2: Reward Modeling (RM)

The Reward Model (RM) learns to score responses according to human preference. Each training entry contains a prompt and three ranked responses; pairwise comparisons are constructed by pairing the best response against each lower-ranked one. The RM is trained by minimizing a pairwise ranking loss:

$$\mathcal{L}_{\text{RM}} = -E_{(x,y_w,y_l)} \left[\log \sigma \left(r(x, y_w) - r(x, y_l)\right)\right] \tag{1}$$

where x is the prompt, y_w and y_l are the preferred and less-preferred responses, $r(\cdot)$ is the predicted reward score, and σ is the sigmoid function.

3.3 Stage 3: Proximal Policy Optimization (PPO)

The SFT model is further trained as a policy using RL, with the RM providing reward signals. To prevent reward hacking—where the model exploits the reward function at the cost of output quality—a KL divergence penalty is added:

$$\mathcal{J}(\theta) = E\left[r(x,y) - \beta \cdot \mathrm{KL}\left(\pi_\theta(\cdot|x) \,\|\, \pi_{\mathrm{SFT}}(\cdot|x)\right)\right] \tag{2}$$

where π_θ is the policy being optimized, π_{SFT} is the frozen SFT reference, and β controls how far the new policy can deviate. A separate Critic model estimates expected rewards to stabilize gradient updates [3].

4 Experiments and Results

4.1 Experimental Setup

Experiments are conducted on the KoChatGPT dataset using KoGPT-2 as the base model. Table 1 summarizes the configuration.

Table 1: Summary of experimental configuration.

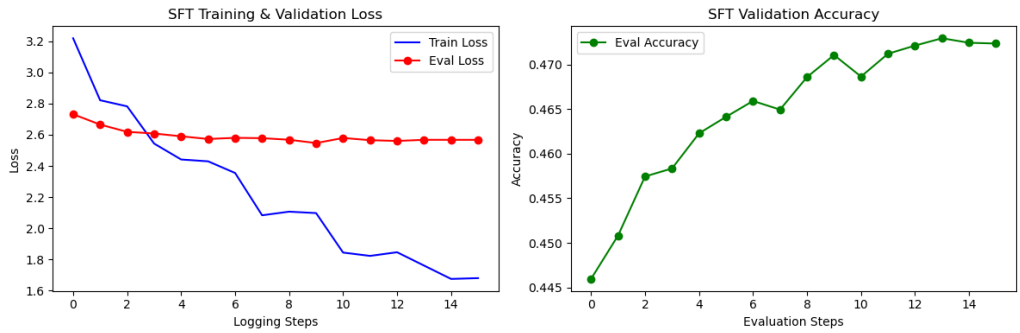
Component	**Details**
Base Model	KoGPT-2 (`skt/kogpt2-base-v2`)
Hardware	T4 GPU
Dataset	KoChatGPT (SFT / RM / PPO splits)
Augmentation	Word2Vec + Beam search + Noise injection
Optimizer	AdamW (SFT) / Adam (RM, PPO)
RL Algorithm	PPO
Evaluation Metrics	BLEU, ROUGE-L, BERTScore (F1), Distinct-1, Perplexity

Three model variants are compared: **SFT-base** (original data), **SFT-aug** (augmented data), and **PPO** (SFT-base further trained via PPO).

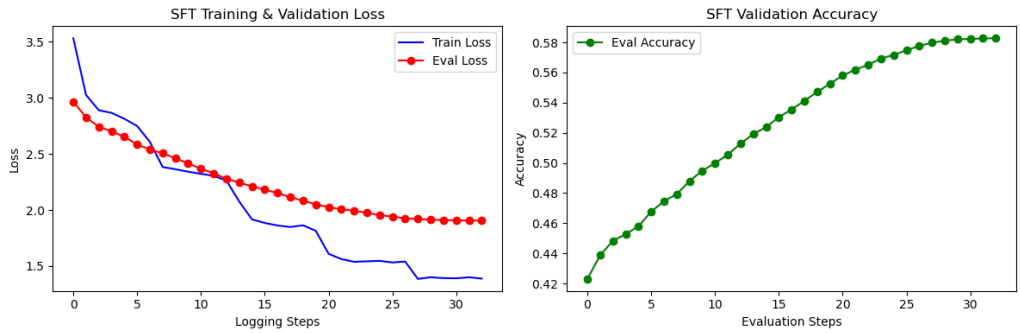
4.2 Evaluation Metrics

Five automatic metrics are used: **BLEU** (n-gram overlap with reference), **ROUGE-L** (longest common subsequence), **BERTScore F1** (semantic similarity via pretrained BERT), **Distinct-1** (proportion of unique unigrams, reflecting lexical diversity), and **Perplexity** (model uncertainty; lower values correspond to more fluent outputs).

4.3 SFT Training Results



(a) SFT-base: eval loss stabilizes at 2.58, accuracy reaches 47.2%.



(b) SFT-aug: eval loss converges to 1.92, accuracy reaches 58.0%.

Figure 2: SFT training dynamics. The augmented model (bottom) converges faster and achieves lower validation loss and higher accuracy than the base model (top).

4.4 Quantitative Evaluation

Table 2: Automatic evaluation results. Best scores per metric are in **bold**.

Model	BLEU	ROUGE-L	BERTScore (F1)	Distinct-1	Perplexity
SFT-base	0.0000	0.0203	**0.6815**	0.6209	7.8109
SFT-aug	0.0000	**0.0316**	0.6649	**0.7173**	8.5531
PPO	0.0000	0.0222	0.6580	0.6757	**7.3946**

BLEU is zero across all models. Zero BLEU is not surprising here—open-ended Korean chatbot responses are naturally varied, and word-for-word overlap with a fixed reference is unlikely by design.

SFT-aug leads on ROUGE-L (0.0316) and Distinct-1 (0.7173). The higher Distinct-1 score in particular shows that augmentation produces more lexically diverse outputs.

SFT-base has the highest BERTScore (0.6815). One possible explanation is that the original training data is closer in wording to the test references, while the augmented and PPO models diverge in phrasing despite being semantically reasonable.

PPO achieves the lowest perplexity (7.3946). The drop from SFT-base (7.8109) points to reward-guided training nudging the model toward more fluent and consistent outputs.

4.5 Qualitative Analysis

Table 3: Selected response comparisons across model variants (English translations provided for readability). Korean model outputs are reproduced verbatim.

Prompt	SFT-base	SFT-aug	PPO
오늘 점심 메뉴 좀 추천해줘. 매콤한 게 당기네. (*Recommend spicy lunch*)	저는 AI챗봇이기 때문에 음식을 먹을 수 없습니다. 하지만 매콤하고 맛있는… (*Refuses; says it cannot eat*)	저는 인공지능 챗봇이기 때문에 식사를 할 수 없습니다. 하지만 매콤한 거 좋아하시… (*Refuses; says it cannot eat*)	As an AI language model, I cannot provide phy... (*Refuses in English —language-switching failure*)
인공지능이 발전하면 인간의 일자리를 모두 대체하게 될까? (*Will AI replace all jobs?*)	저는 AI 챗봇이기 때문에 인간의 일자리를 대체하는 것은 불가합니다. (*Refuses; stays evasive*)	저는 인공지능 어시스턴트이기 때문에 인간의 일자리 대체 문제에 대한 정확한 답변을… (*Gives partial answer*)	저는 인공지능 챗봇이기 때문에 인간의 일자리를 대체하는 것은 불가능합니다. 하지만… (*Concedes limitation, then explains*)
행복은 돈으로 살 수 있다고 생각해? (*Can happiness be bought?*)	저는 AI 어시스턴트이기 때문에 행복에 대한 주관적인 판단을 할 수 없… (*Refuses to give opinion*)	행복은 돈으로도 살 수 있는 것이조. 하지만 행복을 위해서는 노력과 희생이… (*Attempts a balanced answer; minor grammatical error in original output*)	행복하게 살려면 다음과 같은 방법을 시도해 볼 수 있습니다.\n\n1. 목표를 설… (*Provides practical suggestions*)

Across all three prompts, SFT-base defaults to refusing by citing its AI status. SFT-aug shows more varied outputs, though the refusal pattern persists—and occasionally produces minor grammatical errors (e.g., "것이조" for "것이죠"), suggesting the augmented data introduced some noise. The PPO model engages most directly—offering concrete suggestions for the happiness prompt rather than declining. One notable failure: the PPO model responds in English to a Korean input, a language-switching failure that points to language consistency not being enforced during reward modeling.

Table 4: Reward Model scores for SFT-aug and PPO outputs, and the difference (PPO − SFT-aug). Positive values indicate PPO was scored higher by the Reward Model.

#	Prompt (summary)	RM Score (SFT-aug)	RM Score (PPO)	Improvement
0	Spicy lunch recommendation	-3.5737	-3.4919	$+0.0818$
1	Tips for cold weather	-3.5086	-3.5086	0.0000
2	Free Netflix movies	-3.6422	-3.6436	-0.0014
3	Tips for waking up early	-3.3313	-3.3161	$+0.0152$
4	Motivation for gym after work	-3.4875	-3.4436	$+0.0439$
5	Will AI replace human jobs?	-2.7453	-2.8133	-0.0680
6	Mindset for facing challenges	-3.2629	-3.2841	-0.0213
7	Best way to protect the env.	-3.3883	-3.3889	-0.0005
8	Can happiness be bought?	-3.5284	-3.4489	$+0.0795$

PPO scores higher than SFT-aug on five of nine prompts, with the largest gains on prompts 0 and 8 ($+0.08$ each). The three cases where PPO scores lower (prompts 5, 6, 7) involve complex reasoning or open-ended opinion questions, where the reward model may struggle to distinguish better responses. Overall, the trend is consistent with RLHF pushing the model in the right direction.

5 Conclusion and Discussion

This paper implemented and evaluated a full RLHF pipeline for Korean instruction-following, comparing three model variants—SFT-base, SFT-aug, and PPO—on both automatic metrics and qualitative response analysis. Data augmentation at the SFT stage improved validation loss, token-level accuracy, and response diversity (Distinct-1). The PPO model achieved the lowest perplexity, consistent with reward-guided training producing more fluent outputs. One clear failure case—the PPO model responding in English to a Korean prompt—shows that language consistency needs to be explicitly built into the reward signal, and is a natural target for future work.

The reward score gap between SFT-aug and PPO was modest, partly due to the small dataset and lightweight base model. Automatic metrics also have limits as proxies for user preference; human evaluation would give a clearer picture of alignment quality. Nevertheless, the results confirm that a full Korean RLHF pipeline is buildable with open-source tools and limited resources, and should scale as larger models and datasets become available.

References

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